Bias Detection Model Using Graph-Based Preference Analysis

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Abstract

In this project, we tried to build a system that detects whether a person shows bias in their decisions. The idea came from observing that people often choose differently when the same situation is framed in different ways. We modelled user choices as a directed preference graph and used graph analysis along with entropy-based probability to compute a normalized bias score between 0 and 1. The system runs as a Flask web application and uses NetworkX for graph construction and Plotly for visualization.

***Keywords*-preference bias, directed graph, entropy, decision-making, Flask, NetworkX**

I. Introduction

We got the idea for this project while discussing how people make decisions in real life. For example, if you ask someone whether they prefer a high-paying stressful job or a low-paying relaxed one, they might say the relaxed one. But when the same options are framed differently, the same person might flip their choice. That kind of inconsistency is what we are calling preference bias in this project.

We wanted to measure this, not by asking people if they are biased, because nobody really admits that, but by watching their choices across multiple questions and finding the inconsistencies ourselves. The system shows the user a set of pairwise comparison questions and then analyses their answers through a directed graph to see how consistent or skewed their preferences are.

The main contributions of this project are:

- An interactive questionnaire for pairwise preference collection
- A directed weighted graph built from user responses using NetworkX
- A bias scoring formula: Bias = 0.6I + 0.4S
- A Flask web app with Plotly-based graph visualization.

II. Literature Survey

Classical decision theory assumes that people always make rational and transitive choices. If you prefer A over B and B over C, you should prefer A over C. This sounds reasonable but in practice people violate it quite often, especially when options are rephrased or reordered.

Kahneman and Tversky showed through their research that human decisions are heavily influenced by framing, context, and loss aversion. Their work made it clear that bias in decision-making is not just an individual flaw but a systematic pattern in human cognition. Graph-based models have been used in voting theory to detect cycles in collective preferences, which signal contradictions. Shannon entropy is widely used in information theory to measure uncertainty in probability distributions. Our project brings both of these ideas together into a single bias detection framework, which to our knowledge has not been done in this exact form before.

III. Proposed Work

We divided the project into three main parts: collecting user preferences, constructing the graph, and computing the bias score.

A. Questionnaire Interface: We built a web page where users are shown two options at a time and asked to pick one. Questions are based on job scenarios with varying attributes like salary, stress, and flexibility. Each answer is sent to the backend and stored for graph construction.

B. Graph Construction: On the backend, user responses are turned into a directed weighted graph using NetworkX. Each option is a node. Choosing Option A over Option B creates a directed edge from A to B. Repeated preferences increase the edge weight, so stronger or more consistent preferences are reflected in the graph structure.

C. Bias Score: Two values are computed. Inconsistency score I is the ratio of contradictory choice pairs (A over B in one question, B over A in another) to total possible pairs. For each option i, selection probability p_i is computed and Shannon entropy $H = -\Sigma\, p_i \log_2(p_i)$ is calculated and normalized. Skew $S = 1 - H\_norm$. The final score is $Bias = 0.6I + 0.4S$, ranging from 0 (consistent) to 1 (highly biased).

D. Visualization: Results are shown as an interactive Plotly graph. Node size reflects how often an option was chosen, and edge colour reflects preference strength, making it easy to spot dominant choices visually.

The project structure is as follows:

- backend/app.py — Flask server and API routes
- backend/bias_logic.py — I, S, and Bias computation
- backend/graph_builder.py — NetworkX graph construction
- frontend/templates/ — Questionnaire and results HTML
- frontend/static/ — CSS and JavaScript files

IV. Results and Discussion

We tested the system using simulated response sets covering different preference patterns. Users with fully consistent preferences scored between 0.1 and 0.2. Completely contradictory or one-sided responses produced scores near 0.9 to 1.0. Mixed cases fell appropriately in between, which is the behavior we were aiming for.

One thing we noticed is that with fewer questions, even one or two contradictions can push the inconsistency score significantly higher. This means the questionnaire needs enough questions to produce a reliable signal. We also found that question wording matters a lot. Two questions that are essentially asking the same thing in different words can make a consistent user appear inconsistent to the system. This is something we would want to fix in a more polished version.

Overall the scoring worked the way we intended. The graph visualization also helped make the results more understandable, especially for someone who does not have a technical background.

V. Future Work

- Replace the manual scoring weights with values learned from real behavioural data
- Add a live graph that updates as the user answers each question
- Build domain-specific question sets for hiring, lending, and medical decisions
- Deploy as a public web service to support larger behavioural studies
- Implement idealistic bias calculation formulas for larger sample size

VI. Conclusion

This project started from a simple observation that people are not always consistent in their choices and turned into a working system that can actually measure and score that inconsistency. By combining graph theory with entropy-based probability, we built a bias detection tool that is interpretable, does not need labelled training data, and works across different question domains.

We think with better question design and data-driven calibration of the scoring weights, this could become a genuinely useful tool for behavioural research and AI fairness evaluation. It was a good learning experience and one of the more practically interesting projects we have worked on.

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